

RESEARCH PROJECT PROPOSAL

MicroFin-XAI

*A Context-Aware, Explainable and Counterfactual AI Framework for Evidence-Grounded,
Globally Generalizable Financial-Inclusion Interventions*

Project Vision

MicroFin-XAI is designed to answer a question that neither conventional empirical microfinance research nor generic large-language-model financial assistants currently answer on their own: given an individual's measurable financial, behavioral and digital state, which barriers to financial inclusion are globally stable and which are context-dependent, and how can that evidence be converted into a personalized, verifiable intervention rather than a plausible-sounding recommendation? The framework treats machine learning, explainability, counterfactual reasoning, retrieval and language generation as a controlled evidentiary pipeline, not as an assembly of fashionable components, and grounds its central empirical study in a globally representative, publicly available dataset so that the research is reproducible, comparable across contexts, and defensible for a high-impact-factor (Q1) journal submission.

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Research Collaboration Proposal

Interdisciplinary collaboration between Financial Inclusion / Behavioral-Finance Research and AI/ML & Generative AI Research

Document Roadmap

This proposal is organized into thirty-two sections spanning motivation, theory, dataset strategy, system architecture, experimental design, evaluation, ethics and publication planning. Sections 1–9 establish the scientific problem and theoretical grounding; Sections 10–14 define the dataset strategy, including the newly added global public-dataset landscape; Sections 15–21 specify the technical architecture; Sections 22–27 specify the experimental protocol, baselines and evaluation; Sections 28–32 address risk, reproducibility, publication strategy and collaboration.

1. Executive Summary

Financial inclusion research has produced a large body of population-level empirical findings, and machine-learning research has independently produced strong predictive models for credit risk and inclusion status. Neither tradition, on its own, answers a more difficult and more publishable question: for a specific individual, in a specific socioeconomic and geographic context, which barriers actually explain their financial-inclusion state, and what personalized, evidence-grounded action could plausibly change it, without overstating what the data and the model can support?

MicroFin-XAI is a research framework built to answer that question rigorously. It combines predictive machine learning, explainable AI, model-based counterfactual and (where identifiable) causal analysis, multi-strategy evidence retrieval, and a constrained large language model, arranged so that quantitative estimation always precedes language generation. Generative AI is deliberately not treated as a source of truth: client features, calibrated predictions, faithfulness-checked explanations, counterfactual estimates and retrieved evidence form the evidentiary layer, and a verification module controls the language layer's claims, causal wording, and safety boundaries.

The central methodological shift in this revised proposal, relative to the original draft, is the move from a single-country, access-dependent microfinance dataset toward a primary empirical study built on the Global Findex Database 2025 — the World Bank's individual-level, nationally representative survey of financial and digital-financial behavior across 141 economies, collected in 2024 and publicly available through the World Bank Microdata Library. This single change resolves the largest risk in the original proposal (uncertain access to a restricted client-level dataset) while simultaneously creating a substantially stronger and more novel research question: cross-country and cross-context generalization of financial-inclusion drivers, explanations and interventions under distribution shift. A restricted or longitudinal microfinance panel, if access can later be secured, is retained as a secondary track for causal validation rather than as the sole empirical foundation.

Primary Research Question

Which financial-inclusion barriers are stable across countries, regions and socioeconomic contexts, which are context-dependent, and can an explainable, evidence-grounded AI framework translate individual-level predictive and counterfactual evidence into personalized, verifiable interventions without overstating causal claims?

2. Problem Statement

People with superficially similar financial products or opportunities differ substantially in financial knowledge, digital capability, self-efficacy, social capital, financial behavior, and the surrounding institutional and infrastructural context. Two adults with a bank account in different countries — or in different regions of the same country — can face entirely different practical barriers to using that account productively.

Existing empirical research on financial inclusion and microfinance provides valuable population-level relationships (e.g., digital access predicts account usage; financial literacy predicts savings behavior), but a population-level coefficient does not tell an individual what to do next. Conventional machine-learning systems for credit scoring or inclusion prediction go one step further by producing individual predictions, but a

prediction alone is not an explanation, and an explanation alone is not an intervention. Generic large-language-model financial assistants can produce fluent, personalized-sounding text, but that fluency is not evidence: without a quantitative client-state model and a constrained evidence layer, an LLM's advice is neither reliably grounded in the individual's actual measured state nor verifiably supported by financial-inclusion evidence.

A research system is therefore needed that can, in sequence: (1) predict an individual's financial-inclusion state from measurable financial, behavioral, digital and contextual variables; (2) explain why the model reached that prediction, and test whether that explanation is stable across geographic and socioeconomic contexts; (3) estimate plausible intervention effects using methods appropriate to the causal identifiability of the data; (4) retrieve supporting evidence from the financial-inclusion and financial-literacy literature; and (5) communicate the resulting intervention in a personalized, uncertainty-aware, and verifiable form.

3. Literature Positioning and Research Gap

Between 2024 and 2026, several research streams relevant to this proposal have advanced quickly but largely in isolation from one another: (i) machine-learning prediction of microfinance and credit outcomes, including recent rural Bangladesh borrower studies; (ii) explainable ML (SHAP, EBM) for financial-inclusion and microfinance risk assessment; (iii) causal and Double-Machine-Learning estimation of microfinance program effects; (iv) LLM-based financial literacy, financial education and personal-finance assistance; and (v) LLM-plus-ML architectures for credit scoring that extract narrative features for downstream scoring models.

Table 1. Comparison of related research streams against the proposed framework

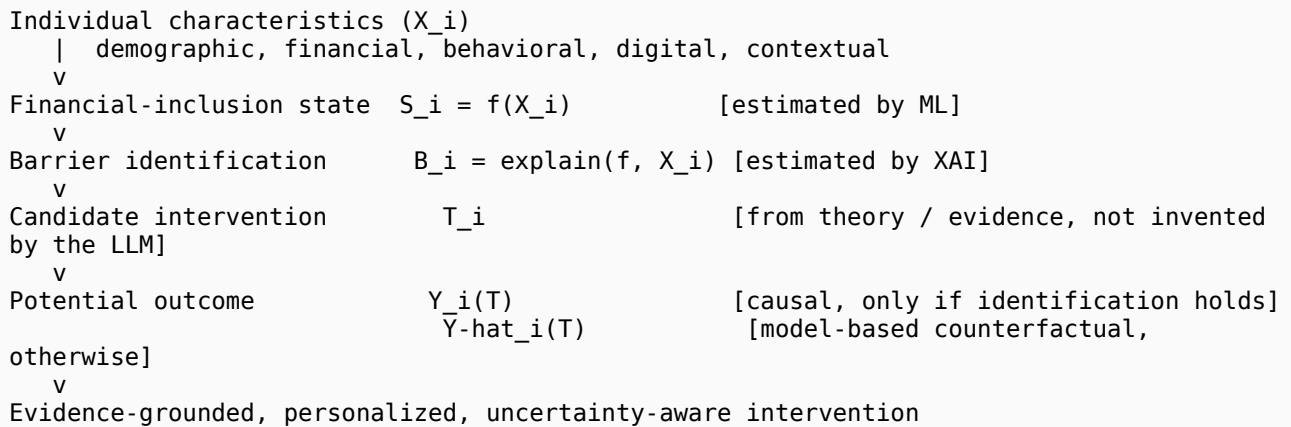
Research Stream	Predict.	XAI	Causal / CF	Cross-Context Generaliz.	LLM	Personalized Interv.
Microfinance + DML (2025)	✓	—	✓	—	—	—
Microfinance / credit-risk ML	✓	—	—	—	—	—
Rural Bangladesh borrower ML (2026)	✓	—	—	—	—	—
Microfinance EBM + SHAP	✓	✓	—	—	—	—
Financial-inclusion XGBoost + SHAP	✓	✓	—	—	—	—
LLM financial literacy / education	—	—	—	—	✓	✓
LLM + credit scoring (narrative features)	✓	—	—	—	✓	—
Cross-country Findex econometric panels	—	—	—	✓	—	—
Proposed: MicroFin-XAI	✓	✓	✓	✓	✓	✓

Key implications for positioning

- Prediction alone — including strong XGBoost/LightGBM baselines already reported for financial inclusion and rural microfinance — is established territory and cannot by itself carry a Q1 contribution.
- SHAP/EBM explainability for financial inclusion is likewise established and should be treated as a rigorous baseline layer, not the primary novelty claim.
- Double Machine Learning has been applied to estimate causal effects of microfinance "plus" services; this work can extend that logic toward individual-level, uncertainty-aware counterfactual reasoning, while being explicit about when causal identification is and is not justified.
- LLM-based financial literacy and personal-finance assistance is an active but largely ungrounded stream: it rarely conditions generation on a quantitative, individual-level, cross-country-validated client-state model.
- Existing cross-country Findex-based econometric work analyzes population-level or country-level relationships; it does not typically test individual-level model and explanation generalization under country/region distribution shift, nor connect that analysis to a personalized intervention-generation pipeline.
- The genuine, defensible novelty of MicroFin-XAI is therefore the combination of (a) a global, individual-level, reproducible empirical study of cross-context generalization in financial-inclusion prediction and explanation, with (b) a constrained, verifiable evidence-to-language pipeline that turns that analysis into personalized, uncertainty-aware interventions.

4. Theoretical Framework

This section, newly formalized in the present revision, defines the conceptual model that the empirical study is designed to test. It replaces an implicit architecture-first framing with an explicit individual-state-to-outcome model.



Formally, the individual is represented by a feature vector X_i spanning demographic, financial, behavioral, digital and contextual (country/region/income-group) variables. A predictive model estimates the financial-inclusion state $S_i = f(X_i)$. An explainability layer decomposes f into per-feature contributions to identify candidate barriers B_i . A candidate intervention T_i is drawn from a theoretically and empirically justified intervention taxonomy — never invented ad hoc by the language model. The effect of that intervention is expressed as a potential outcome $Y_i(T)$; this quantity is only interpreted causally when a formal identification

check (Section 18) is satisfied, and is otherwise explicitly reported as a model-based counterfactual, $\hat{Y}_i(T)$, with that distinction preserved through to the final generated text.

A second, equally important theoretical addition is the treatment of context itself as a variable of scientific interest rather than a nuisance factor to be pooled away. Country, region, income group, gender, age cohort and rural/urban residence are modeled explicitly as conditioning variables, so that the framework can distinguish globally stable predictors of financial inclusion from predictors whose importance is context-dependent. This distinction — stable versus context-dependent barriers — is the paper's central empirical contribution and is only testable with a dataset that has genuine cross-country and cross-context coverage, which is the primary reason for repositioning the dataset strategy around the Global Findex Database (Section 10).

5. Research Questions

1. Which financial, behavioral, digital and contextual variables predict individual-level financial-inclusion status, and how does predictive performance compare against established gradient-boosting baselines?
2. Which predictors are globally stable across countries, regions and income groups, and which are context-dependent?
3. Do model performance and explanation patterns degrade in a measurable, characterizable way under country-held-out, region-held-out and income-group-held-out evaluation (i.e., under distribution shift)?
4. Can explainable ML produce explanations that are faithful to model behavior and stable enough to support intervention reasoning across heterogeneous contexts?
5. Under what conditions can counterfactual or causal methods identify plausible intervention pathways, and when must the framework fall back to explicitly labeled model-based counterfactuals?
6. How do dense, hybrid and graph-based retrieval strategies differ in evidence-retrieval and generation quality for financial-inclusion questions, and is the added complexity of graph-based retrieval justified by measurable gains?
7. Does grounding LLM-generated interventions in validated ML, explanation and counterfactual outputs improve personalization, factual support and expert acceptability relative to ungrounded generation?
8. Which components of the integrated architecture are individually necessary to achieve reliable, evidence-grounded intervention quality, and which add cost without measurable benefit?

6. Research Hypotheses

The single broad hypothesis in the original proposal is decomposed here into eight modular, independently testable hypotheses, each corresponding to a specific experiment described in Sections 22–26.

H1 — Predictive representation: Financial, behavioral, digital and contextual variables jointly provide significant, complementary predictive information for individual-level financial-inclusion status, beyond any single variable group alone.

H2 — Context dependence: The relative importance of financial-inclusion predictors varies systematically across countries, regions, income groups and demographic subgroups.

H3 — Generalization under distribution shift: Model performance and explanation patterns show measurable, structured degradation when evaluated on held-out countries, regions or income groups, relative to random-split evaluation.

H4 — Explanation faithfulness and stability: Faithfulness-validated explanations are more consistent across contexts and more useful for intervention reasoning than unvalidated or purely correlational feature-importance rankings.

H5 — Counterfactual actionability: Constrained, plausibility-checked counterfactual reasoning produces intervention pathways rated as more actionable than unconstrained free-form recommendation generation.

H6 — Evidence grounding: Retrieval-grounded generation reduces unsupported claims and improves factual/source support in generated interventions relative to ungrounded LLM generation.

H7 — Personalization: Interventions conditioned on individual-level predictive, explanatory and counterfactual outputs are rated more relevant by domain experts than generic financial-literacy content.

H8 — Verification and calibrated abstention: Claim-level verification combined with uncertainty-based abstention reduces unsupported or overconfident statements without materially reducing the system's useful coverage.

7. Aim and Objectives

Aim: To develop and empirically evaluate a context-aware, explainable and counterfactual AI framework that predicts individual-level financial-inclusion status from globally comparable public data, tests the stability of its predictors and explanations across countries and contexts, and converts validated evidence into personalized, verifiable interventions.

9. Construct an analysis-ready individual-level dataset from the Global Findex Database 2025, with a clearly justified primary outcome variable and a documented variable taxonomy (financial, behavioral, digital, contextual).
10. Benchmark interpretable and gradient-boosting ML models for the primary financial-inclusion prediction task, evaluated with leakage-controlled, group-aware validation.
11. Develop and validate explainability methods (SHAP and complementary techniques) that identify global and individual-level predictive drivers, and test their faithfulness and cross-context stability.
12. Formalize a causal-identification gate that determines, per research question, whether Double-Machine-Learning-style causal estimation is justified or whether the analysis must be reported as a model-based counterfactual.
13. Design and execute a cross-country, cross-region and income-group distribution-shift evaluation protocol to identify stable versus context-dependent predictors.
14. Construct a versioned, source-attributed evidence corpus covering financial inclusion, financial literacy, digital financial behavior and relevant institutional/policy literature.
15. Compare dense, hybrid and graph-based retrieval architectures for evidence grounding, evaluated on both retrieval quality and downstream generation quality.

16. Develop a constrained LLM intervention layer that receives structured client-state, explanation, counterfactual and evidence inputs and cannot alter quantitative outputs or invent causal claims.
17. Implement a claim-level verification and calibrated-abstention layer, and evaluate its effect on unsupported-claim rate and expert-rated reliability.
18. Evaluate the complete framework through predictive, explanatory, retrieval, generation, counterfactual and blinded expert-evaluation metrics, with a full ablation study isolating each component's contribution.

8. Scope and Target Population

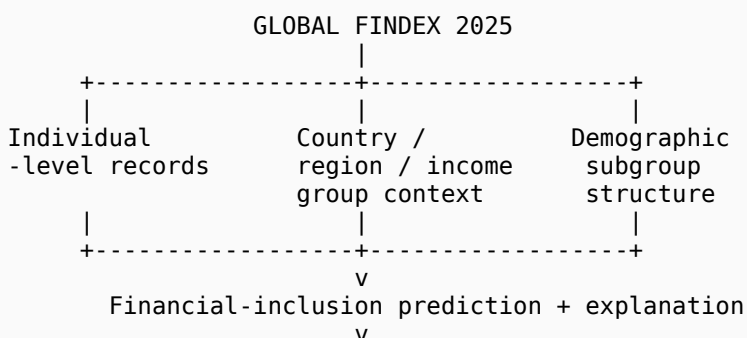
The primary empirical study targets the general adult population covered by the Global Findex Database 2025 — approximately 145,000 individual-level respondents across 141 economies, collected in 2024 — rather than a single-country microfinance client cohort. This broadens the target population from "microfinance borrowers" to "financially included or excluded adults," which is both scientifically defensible (financial inclusion is the theoretically prior construct that microfinance access, usage and effectiveness sit within) and practically necessary for the cross-context generalization design that anchors the paper's novelty.

A single, clearly justified primary outcome variable will be fixed before model development begins (Section 12), to avoid the common failure mode of an unfocused paper reporting many weakly related targets. Candidate primary outcomes, to be finalized with the domain researcher, include formal account ownership and usage, a composite financial-inclusion index constructed from validated Findex sub-indicators, or a financial-resilience indicator. Secondary/microfinance-specific outcomes may be analyzed once a restricted or longitudinal partner dataset (Section 11.3) becomes available, positioned explicitly as a causal-validation track rather than the primary empirical foundation.

The LLM component remains decision-support and financial-education/intervention-support only. It is not authorized to perform autonomous credit decisions, guarantee financial outcomes, or issue individualized regulated financial advice beyond the research scope.

9. Why "Global" Means a Scientific Design, Not a Bigger Dataset

A common and weaker way to use a multi-country dataset is to pool every country into one table and train a single model for marginally higher accuracy. That produces an ordinary cross-country ML paper. MicroFin-XAI instead treats cross-country and cross-context heterogeneity as the object of study.



Stable drivers	vs.	Context-dependent drivers
	v	
Cross-country / cross-region generalization test		

Concretely, this reframes the research questions from "can we predict financial inclusion" (largely already answered in the literature) to four sharper questions: which predictors of financial inclusion hold globally; which are specific to a region, income group, or digital-infrastructure context; how much does predictive and explanatory performance degrade when a model trained on some countries is evaluated on entirely unseen countries; and whether a personalized-intervention system can remain evidence-grounded and appropriately cautious when the client's context shifts away from the training distribution. Each of these is directly testable with Global Findex data and is not adequately addressed by the existing literature streams summarized in Table 1.

10. Dataset Strategy: A Tiered, Access-Realistic Design

The most significant risk identified in reviewing the original proposal was dataset access: the most topically ideal datasets (a 2026 BRAC University rural-Bangladesh borrower dataset; a 3,121-respondent RPDC financial-literacy and access survey; a historical PKSf longitudinal microfinance panel) are scientifically attractive but their public, raw-data accessibility could not be confirmed at the time of review. Building the paper's primary empirical claim on an unconfirmed dataset is a serious risk to a Q1 submission timeline. The dataset strategy is therefore restructured into four explicit tiers, so that the paper's central contribution never depends on an access request that might not be granted.

10.1 Tier 1 — Primary dataset (*public, confirmed, immediately usable*)

Global Findex Database 2025 (World Bank / Gallup World Poll), individual-level microdata, accessed through the World Bank Microdata Library. Key documented characteristics:

- Nationally representative surveys of approximately 145,000 adults across 141 economies, conducted in calendar year 2024, published as the fifth edition of the Global Findex (following the 2011, 2014, 2017 and 2021 editions).
- Individual-level microdata are downloadable, together with the full questionnaire, methodology documentation and codebook, supporting full reproducibility.
- Coverage includes account ownership, savings, borrowing, digital payments, mobile-money use, financial resilience, mobile-phone ownership, internet use and digital safety — spanning financial, digital and behavioral variable families required by the original proposal's variable taxonomy.
- The database additionally reports indicators disaggregated by gender, income tercile/quintile grouping, labor-force participation, age cohort and rural/urban residence, which directly supports the subgroup and context-dependence analyses in Sections 9 and 24.
- A single-country microdata extract (e.g., Bangladesh, survey ID BGD_2024_FINDEX_v02_M) is separately downloadable for any country-specific robustness or case-study analysis the domain researcher wishes to add.

10.2 Tier 2 — Secondary public datasets (*robustness, supplementary constructs*)

These datasets are publicly available but individually too small, too narrow, or too dated to serve as the primary empirical foundation. Each is retained for a specific, limited role.

Dataset	Approx. N	Public access	Role in MicroFin-XAI
Prior Global Findex waves (2011, 2014, 2017, 2021)	~150,000 per wave	Confirmed — World Bank Microdata Library	Temporal robustness / distribution-shift-over-time analysis
Peru financial literacy & inclusion survey (Zenodo, 2024)	118	Confirmed — open Zenodo dataset with questionnaire and codebook	Construct/instrument comparison; supplementary behavioral validation
Morocco digital microfinance & algorithmic bias (Zenodo, 2026)	250	Confirmed — open Zenodo dataset	Supplementary digital-literacy / fairness robustness check
Electronic payment & financial inclusion survey (Zenodo, 2025)	100	Confirmed — open Zenodo dataset	Small-sample external sanity check on digital-payment constructs

10.3 Tier 3 — Access-pending datasets (do not depend on these for the core paper)

These datasets are topically close to the original microfinance framing but their raw-data accessibility was not confirmed during this review. They are retained as an upside, pursued in parallel, but the paper's timeline and core claims must not depend on them.

- Bangladesh rural microfinance borrower dataset (BRAC University, 2026 thesis): 1,001 borrowers, 85 engineered features spanning digital literacy, household infrastructure, lifestyle and psychometric indicators. Repository page describes the dataset; raw-file public download was not verified. Recommended action: contact the author/supervisor to request anonymized research-use access.
- RPDC Financial Literacy & Financial Access Survey (BRAC Business School, data collected November 2021, reported in a September 2026 paper): 3,121 respondents across all eight administrative divisions of Bangladesh, with a validated five-item financial-literacy score and detailed financial-access variables. Access not confirmed from the paper alone.
- PKSF Monitoring and Evaluation Study / Follow-up Monitoring and Evaluation Study: a historical longitudinal panel of more than 2,500 Bangladeshi households across multiple waves (1997–98, 1999/2000, 2004), designed specifically to evaluate microcredit program effects. This is scientifically the most valuable dataset for genuine causal (treatment-to-outcome) identification, but it is old and public raw-data access was not confirmed.

10.4 Tier 4 — Evidence corpus for retrieval (not a training dataset)

The retrieval layer (Section 17) draws on a separate, versioned literature and institutional-report corpus rather than survey microdata: World Bank financial-inclusion publications and the Global Findex report series; CGAP microfinance and financial-inclusion evidence; IMF Financial Access Survey documentation; OECD/INFE financial-literacy frameworks; peer-reviewed financial-inclusion, microfinance, financial-literacy and digital-finance literature (2019–2026); and relevant national policy documents. This corpus is kept strictly separate from evaluation questions to avoid retrieval leakage.

Why this tiering matters for a Q1 timeline

The paper's central predictive, explanatory and cross-context generalization claims (H1–H4, and the stable-vs-context-dependent-driver contribution) are fully supportable using only confirmed, immediately downloadable Tier 1 and Tier 2 data. Tier 3 access requests are pursued in parallel as an opportunity to add a genuine causal-validation track (Section 11.3), but a rejected or delayed access request cannot block submission, because the core paper does not depend on it.

11. Variable and Construct Framework

11.1 Variable taxonomy mapped to Global Index

Construct family	Example variables (Global Index 2025)	Role in framework
Financial	Account ownership, savings behavior, formal/informal borrowing, payments, financial resilience (ability to access emergency funds)	Core predictors and candidate outcome components
Digital	Mobile-phone ownership, internet use, digital-payment use, mobile-money use, digital safety experience	Core predictors; central to the Digital Connectivity Tracker module introduced in the 2025 edition
Behavioral	Frequency of transactions, saving purpose/method, borrowing purpose/source, reported barriers to account ownership	Behavioral-finance layer connecting to financial self-efficacy theory
Demographic / socioeconomic	Gender, age cohort, education, income group (wealthiest 60% vs poorest 40%), labor-force participation, rural/urban residence	Subgroup and context-dependence analysis (Section 9)
Contextual	Country, region, income-group classification, survey mode (face-to-face vs. telephone)	Distribution-shift and generalization experiments (Section 24)

11.2 Primary outcome definition

The primary outcome will be finalized jointly with the domain researcher before model development begins, from among the following theoretically justified candidates, in order of recommended priority: (1) a composite financial-inclusion index constructed from validated ownership/usage sub-indicators; (2) formal account ownership and active usage (a stricter, better-justified target than ownership alone); (3) a financial-resilience indicator based on the ability to raise emergency funds. Whichever is selected, the definition, construction formula, and any imputation rules will be fixed and documented before the evaluation split is created (Section 22), to prevent target leakage and post hoc redefinition.

11.3 Causal-validation track (conditional, Tier 3)

If access to a longitudinal or intervention-based microfinance dataset (Section 10.3) is obtained, it is used exclusively for the causal-validation track: estimating treatment-to-outcome effects of a defined microfinance intervention, under the causal-identification gate in Section 18. This track is additive and clearly separated in the manuscript from the Global-Index-based observational/generalization track, so that the paper never mixes causal claims from one dataset with associational claims from another.

12. Global Public Dataset Landscape for a High-Impact Financial-Inclusion Study

This section is added specifically to document, for the domain researcher and for manuscript reviewers, which publicly available datasets exist worldwide for this research area, what each offers, and why the combination selected in Section 10 is the strongest available basis for a Q1 submission. It also serves as a citable resource-availability statement, which reviewers of data-driven social-science and fintech journals increasingly expect.

12.1 Tier A — Large, individual-level, cross-country (primary-study grade)

Dataset	Coverage	N (individual-level)	Access
Global Findex Database 2025	141 economies, 2024 data	~145,000	Public — World Bank Microdata Library
Global Findex 2021 / 2017 / 2014 / 2011 (prior waves)	Up to ~140+ economies each wave	~140,000–150,000 per wave	Public — World Bank Microdata Library
IMF Financial Access Survey (FAS)	Country-level, 189+ countries	Country-level indicators, not individual microdata	Public — IMF data portal (supply-side, complementary to Findex demand-side data)

12.2 Tier B — National / regional, individual-level (secondary-study grade)

Dataset	Country / region	N	Access	Best use
Bangladesh rural microfinance borrowers (BRAC Univ., 2026)	Bangladesh	1,001 (85 features)	Access request required	Deep single-country microfinance case study
RPDC Financial Literacy & Access Survey	Bangladesh (8 divisions)	3,121	Access request required	Financial-literacy construct validation
PKSF MES / FMES longitudinal panel	Bangladesh (historical)	>2,500 households, multi-wave	Access request required	Causal / longitudinal validation
Morocco digital microfinance & algorithmic bias (2026)	Morocco	250	Open — Zenodo	Digital-literacy and fairness robustness
Peru financial literacy & inclusion (2024)	Peru (Lima)	118	Open — Zenodo	Instrument/construct comparison
Electronic payment & financial inclusion (2025)	Not country-specific in source	100	Open — Zenodo	Small-sample digital-payment sanity check
Africa financial-inclusion panel (2026)	44 African countries, 2010–2021	Country-year panel	Open — Zenodo	Country-level econometric robustness only, not individual-level personalization

12.3 Tier C — Supply-side and institutional context data (not for ML training, used for framing and evidence corpus)

- World Bank / CGAP microfinance and financial-inclusion policy reports and working papers.
- IMF Financial Access Survey (supply-side indicators: number of MFI branches, loan accounts per 1,000 adults, deposit accounts, etc.) — useful for contextualizing demand-side Findex findings at the country level.
- OECD/INFE international financial-literacy survey frameworks and country reports — useful for construct validation of any financial-literacy variables used.
- 60 Decibels Microfinance Index — a repeated cross-country client-outcomes survey series covering 2022–2025, useful as an interpretive reference for microfinance-specific client outcomes even where its microdata is not directly merged into the ML pipeline.

12.4 Selection rationale

Applying the explicit dataset-selection criteria formalized in Section 13 — individual-level observations, cross-country comparability, temporal coverage, outcome availability, geographic diversity and reproducibility/public accessibility — the Global Findex Database 2025 is the only dataset in this landscape that satisfies all six criteria simultaneously and without an unresolved access dependency. This is the core justification for its selection as the Tier 1 primary dataset in Section 10, and it is this global landscape review, rather than convenience, that supports the claim of an access-realistic, reproducible, Q1-appropriate empirical design.

13. Formal Dataset Selection Criteria

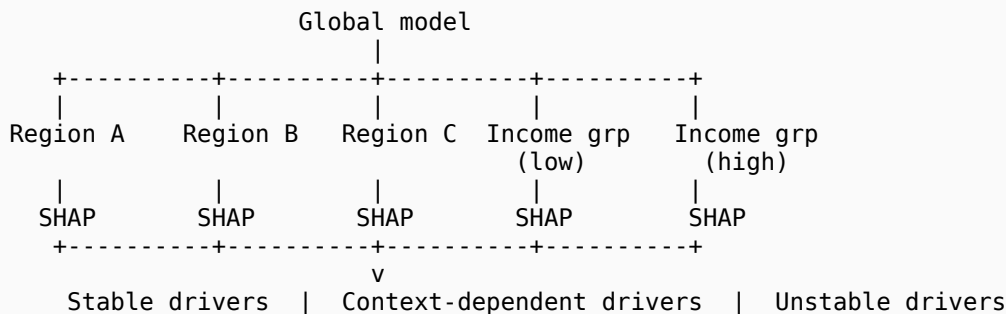
To make the dataset choice a documented methodological decision rather than an ad hoc one, the primary dataset D is required to satisfy the following criteria set, $D = \{I, C, T, O, G, R\}$:

- I — Individual-level observations (not only country aggregates), required for personalized-intervention modeling.
- C — Cross-country comparability, using harmonized questions and definitions, required for the generalization design in Section 9.
- T — Temporal coverage or repeated waves, enabling robustness checks and, where feasible, temporal distribution-shift analysis.
- O — Availability of a theoretically justified, measurable outcome variable.
- G — Genuine geographic and socioeconomic diversity, spanning multiple regions and income groups.
- R — Reproducibility: public accessibility, documented methodology, and a codebook sufficient for independent replication.

The Global Findex Database 2025 satisfies I (individual microdata via the World Bank Microdata Library), C (harmonized questionnaire across 141 economies), T (five waves since 2011), O (multiple validated financial-inclusion and resilience indicators), G (141 economies across all regions and income groups) and R (public microdata, full questionnaire and codebook, documented Gallup World Poll sampling methodology).

14. Global versus Context-Specific Pattern Analysis

This analysis is one of the paper's central empirical contributions. Rather than reporting a single global SHAP ranking, feature importance is estimated conditionally on context and compared.



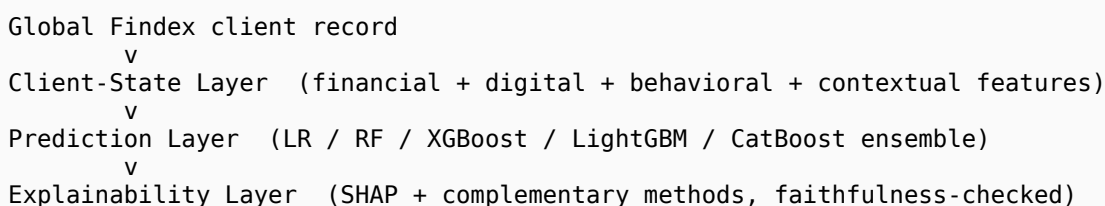
Drivers are classified into three categories: globally stable drivers (consistently high importance and consistent sign across contexts); context-specific drivers (high importance only within particular regions, income groups, or digital-infrastructure contexts); and unstable drivers (importance or sign changes substantially across contexts, flagged for cautious interpretation). This classification, and not raw predictive accuracy, is proposed as the paper's headline empirical finding, because it is the piece of evidence that most directly informs whether a personalized-intervention system can safely transfer conclusions from one context to another — a question of clear interest to financial-inclusion policy audiences and therefore to high-impact-factor development-economics and fintech journals.

15. Representation Stability Analysis

A complementary analysis asks whether the model learns a genuinely transferable representation of financial-inclusion state or merely memorizes country-specific patterns. Proposed analyses include embedding-distance and centroid-similarity comparisons across countries/regions, clustering-consistency checks, and domain-separability tests (e.g., whether a simple classifier can distinguish which country a representation came from — high separability would indicate context leakage rather than a universal representation). This analysis directly operationalizes H3 and gives the generalization claim (Section 9) a concrete, falsifiable measurement rather than a qualitative description.

16. Proposed System Architecture

The architecture is organized as a controlled, evaluable pipeline in which quantitative estimation always precedes language generation, and each stage can be benchmarked independently.



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      v
Context Layer (country / region / income-group conditioning)
      v
Barrier Identification
      v
Causal-Identification Gate --No--> Model-Based Counterfactual (labeled as such)
      |Yes
      v
Causal / DML Estimation
      v
Uncertainty Estimation (prediction / explanation / counterfactual / evidence)
      v
Evidence Retrieval (Dense / Hybrid / GraphRAG)
      v
Constrained LLM Reasoner
      v
Claim-Level Verifier --Fail--> Revise / Abstain / Human review
      |Pass
      v
Personalized, Evidence-Grounded Intervention Report

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Safety boundary

The LLM may recommend interventions and generate explanatory language, but it never changes a quantitative prediction, never invents a causal effect, and never advances a claim up the causal-claim ladder (Section 19) beyond what the analytical layer has certified. All causal or counterfactual estimates originate in the analytical layer and are passed to the LLM strictly as controlled, labeled inputs.

Table 2. Core modules and research value

Module	Function	Research value
Client-State Layer	Normalize structured client and behavioral information from Global Findex records	Produces a reproducible, documented individual representation
Prediction Layer	Compare LR, RF, XGBoost, LightGBM, CatBoost and a selected neural baseline	Quantifies predictive capability against strong published financial-inclusion baselines
Explainability Layer	SHAP plus complementary global/local explanation methods	Links predictions to interpretable, faithfulness-tested factors; treated as a rigorous baseline layer
Context Layer	Conditions analysis on country / region / income group / demographic subgroup	Enables the stable-vs-context-dependent-driver contribution (Section 14)
Counterfactual / Causal Layer	Model-based counterfactuals by default; DML-style causal estimation only when the identification gate (Section 18) is satisfied	Separates causal estimation from LLM language generation; central scientific-integrity safeguard
Evidence Layer	Dense, hybrid and graph-based retrieval over a versioned financial-inclusion corpus	Tests retrieval architecture empirically rather than assuming one method is best
LLM Layer	Converts validated analytical outputs into personalized natural-language	Adds personalization strictly under quantitative and evidentiary

Module	Function	Research value
	interventions	constraints
Verification Layer	Checks evidence support, internal consistency, causal-claim level, and prohibited content	Directly controls hallucination and unsafe or overstated advice
Evaluation Layer	Automated metrics plus blinded expert evaluation	Supplies the statistical evidence for every hypothesis in Section 6

17. Core Technical Modules

17.1 Data and feature engineering

- Financial variables: account ownership, saving, formal/informal borrowing, digital and non-digital payments, financial resilience.
- Digital variables: mobile-phone ownership, internet use, digital-payment use, mobile-money use, digital-safety experience (Digital Connectivity Tracker module of the 2025 Findex edition).
- Behavioral variables: transaction frequency, saving/borrowing purpose and source, self-reported barriers to account ownership.
- Demographic/socioeconomic variables: gender, age cohort, education, income group, labor-force participation, rural/urban residence — included only where theoretically justified and consistent with the ethics framework in Section 27.
- Contextual variables: country, region, income-group classification, and survey-mode indicator (to control for face-to-face versus telephone collection-mode effects, which the Findex methodology documents as varying by country and wave).

17.2 Prediction layer

Candidate models: Logistic Regression as an interpretable statistical baseline; Random Forest; XGBoost; LightGBM; CatBoost; and a compact neural baseline if dataset size and structure justify it. Hyperparameter tuning and cross-validation are performed strictly within training folds, with group-aware (country-aware) splitting used wherever a distribution-shift experiment requires it, to prevent information leakage across the held-out grouping variable. Metrics are selected to match the finalized outcome type: ROC-AUC, PR-AUC, F1, precision, recall, balanced accuracy, log loss, Brier score and calibration error for classification outcomes; MAE, RMSE and R^2 if a continuous financial-resilience index is used.

17.3 Explainability layer

SHAP provides global and local explanations for tree-based models; permutation importance, partial dependence and accumulated local effects are used as complementary, model-agnostic checks. Explanation faithfulness is evaluated directly (e.g., via deletion/insertion or perturbation-based faithfulness tests) rather than assumed. Because SHAP-based explanation for financial inclusion is established in the literature, this layer is explicitly positioned as a necessary, rigorously validated baseline component, with the paper's novelty resting on the cross-context stability analysis built on top of it (Section 14), not on the explanation method itself.

17.4 Counterfactual / causal layer

By default, the framework reports model-based counterfactuals: re-scoring a client's predicted state under a hypothesized change to a modifiable feature (e.g., digital-payment adoption), validated against sparsity, proximity and domain-plausibility constraints so that generated counterfactuals correspond to realistic, actionable changes rather than arbitrary feature perturbations. Where the causal-identification gate (Section 18) is satisfied — which will typically require the Tier 3 longitudinal/intervention dataset described in Section 11.3, since the cross-sectional Global Findex survey cannot by itself support strong causal identification of a specific intervention's effect — Double-Machine-Learning-style estimation is used instead, and the distinction between the two is preserved through every downstream stage, including the final generated text.

17.5 Evidence retrieval layer

A curated, versioned evidence corpus (Section 10.4) supports three retrieval families, compared empirically rather than assumed superior: dense vector retrieval; hybrid lexical-plus-dense retrieval; and graph-enhanced (GraphRAG) retrieval. A consistent reranking step is applied where appropriate across all three, so that comparisons isolate the retrieval mechanism rather than confounding it with reranking differences. The corpus is kept separate from evaluation questions to prevent retrieval leakage.

17.6 LLM intervention layer

The LLM receives a structured client-state summary, model prediction with calibrated uncertainty, validated explanation, counterfactual or causal estimate (correctly labeled), and retrieved evidence. It produces a personalized intervention containing the identified barrier, a plain-language explanation, recommended educational or action steps, evidence references, stated uncertainty/limitations, and escalation conditions. The LLM cannot alter the quantitative prediction or invent a causal effect; it is strictly a controlled synthesis and communication layer, consistent with the safety boundary in Section 16.

17.7 Verification and safety layer

- Evidence-support checker — each substantive factual claim is mapped to retrieved evidence where feasible; unsupported claims are flagged.
- Consistency checker — generated text is checked against the structured ML/counterfactual output for contradiction.
- Causal-claim-level checker — generated language is checked against the causal-claim ladder (Section 19) so that no statement claims a stronger evidentiary status than the analytical layer certified.
- Uncertainty checker — statements not accompanied by appropriate uncertainty qualifiers, where the underlying estimate is uncertain, are flagged or revised.
- Safety checker — prohibits autonomous credit-approval/denial language, guaranteed-outcome language, and individualized regulated financial advice beyond the study's decision-support scope.
- Human-review flag — high-uncertainty or high-impact cases are routed to a qualified reviewer rather than returned as a final recommendation.

18. The Causal-Identification Gate

This is a formal safeguard, elevated in this revision to its own section given its importance to the scientific integrity of the paper. Before any effect estimate is reported as causal, the following sequential checks must all be satisfied; failing any check routes the analysis to the model-based-counterfactual branch instead.

```
Treatment (intervention) clearly defined? --No--> Model-based counterfactual
|Yes
Outcome clearly defined and measured post-treatment? --No--> Model-based counterfactual
|Yes
Temporal ordering (treatment precedes outcome) established? --No--> Model-based
counterfactual
|Yes
Relevant confounders measured? --No--> Model-based counterfactual
|Yes
Overlap / positivity holds across treatment groups? --No--> Model-based counterfactual
|Yes
Consistency assumption plausible? --No--> Model-based counterfactual
|Yes
Conditional exchangeability plausible, with stated justification? --No--> Model-based
counterfactual
|Yes
v
Causal estimation permitted (e.g., DML), with assumptions reported in full
```

Because the Global Findex Database is a cross-sectional demand-side survey rather than a randomized or longitudinal intervention study, causal estimation of a specific microfinance intervention's effect is not expected to pass this gate using Tier 1 data alone. This is stated explicitly and is not treated as a weakness to be hidden: the paper's Global Findex track is framed throughout as an observational, associational and generalization study, and any causal claim is reserved for the separate Tier 3 causal-validation track (Section 11.3), contingent on securing an appropriate longitudinal or intervention dataset.

19. The Causal-Claim Ladder

To keep every generated statement calibrated to its actual evidentiary status, all claims — human-written or LLM-generated — are classified on a five-level ladder. The LLM and the verification layer are both constrained so that no statement is worded at a higher rung than the analytical layer has certified for that specific claim.

Level	Evidentiary status	Permitted phrasing (example)
1	Association	"X is associated with Y in this population."
2	Predictive contribution	"X improves the model's prediction of Y."
3	Model-based counterfactual	"If X were changed in the model, predicted Y would change by an estimated amount, under stated assumptions."
4	Causal estimate	"Under the stated identification assumptions, intervention T is estimated to change outcome Y."
5	Individual treatment effect	"For this individual, under valid identification, the estimated effect of T on Y is [value], with uncertainty [interval]."

20. Uncertainty Quantification and Calibrated Abstention

Uncertainty is tracked and propagated at four distinct levels, rather than collapsed into a single confidence score: (1) prediction uncertainty from the ML model; (2) explanation uncertainty, reflecting the stability of SHAP or other attribution values under resampling or perturbation; (3) counterfactual/causal uncertainty, reflecting the reliability of the estimated intervention effect; and (4) evidence uncertainty, reflecting the strength and consistency of retrieved supporting evidence.

A composite uncertainty score U combines these four levels through a documented aggregation rule, and drives a calibrated abstention decision:

Decision =	Recommend	if $U < \text{threshold}$
	Qualified recommend	if $\text{threshold} \leq U < \text{high-threshold}$
	Human review	if $U \geq \text{high-threshold}$

The abstention mechanism is evaluated using selective-prediction metrics: coverage (the proportion of cases the system is willing to address directly), selective risk (error rate among accepted cases), and calibration of U against observed error. This operationalizes H8 and directly addresses the practical and ethical concern that an AI system should recognize the limits of its own evidence rather than generating a plausible-sounding recommendation for every case.

21. Expert and Blinded Evaluation Protocol

Domain experts (financial-inclusion researchers and, where feasible, microfinance/financial-education practitioners) independently score system outputs, blinded to which system configuration generated each output, across: factual correctness, evidence adequacy, personalization, actionability, plausibility, safety, causal-claim appropriateness (i.e., whether the wording matches the correct rung of the ladder in Section 19), and overall usefulness.

Systems compared: LLM-only (no quantitative grounding); ML + LLM (quantitative state, no explanation); ML + XAI + LLM (adds explanation); ML + XAI + Counterfactual + LLM (adds counterfactual reasoning); and the full MicroFin-XAI system with retrieval and verification. Inter-rater reliability, paired statistical tests (e.g., paired Wilcoxon or bootstrap comparison across system configurations) and effect sizes are reported for every comparison, with confidence intervals rather than point estimates alone.

22. Experimental Methodology

The study follows a leakage-controlled protocol. The dataset is divided into training, validation and held-out test partitions, using group-aware (country-aware) splitting wherever a distribution-shift experiment requires it. All preprocessing and feature-selection steps that can learn from data are fitted only on training folds.

19. Fix the primary outcome definition and theoretical construct model jointly with the domain researcher, before any model development.
20. Perform data-quality analysis: missingness patterns, outlier analysis, and leakage screening specific to survey-weighted data.
21. Create a locked evaluation split (random, country-held-out, region-held-out and income-group-held-out variants) before extensive model development begins.

22. Train and tune ML baselines using nested or appropriately separated validation, applying Global Findex survey weights correctly throughout.
23. Fit the explainability layer on the selected model(s) and run faithfulness and cross-context stability tests.
24. Run the causal-identification gate (Section 18) for any proposed intervention effect; branch to causal or counterfactual estimation accordingly.
25. Build and index the evidence corpus (Section 10.4); run retrieval-quality evaluation for dense, hybrid and graph retrieval.
26. Generate interventions through the constrained LLM layer, apply the verification layer, and log claim-level outcomes.
27. Conduct the blinded expert-evaluation study (Section 21).
28. Conduct the full ablation study (Section 23) and the cross-context generalization experiments (Section 24).
29. Conduct failure analysis (Section 25) and finalize statistical reporting (Section 26).

23. Component-Necessity (Ablation) Study

Rather than presenting the complete architecture and assuming every component is beneficial, component necessity is tested directly through an incremental ablation sequence, evaluated on identical held-out data at every step.

Configuration	Components included
S0	LLM only (no quantitative grounding)
S1	ML prediction only
S2	ML + Explainability (XAI)
S3	ML + XAI + Counterfactual / causal layer
S4	ML + XAI + Counterfactual + Evidence retrieval
S5	ML + XAI + Counterfactual + Evidence retrieval + LLM
S6	S5 + Claim-level verification
S7 (full system)	S6 + Uncertainty quantification and calibrated abstention

This directly answers Research Question 8 and provides the evidentiary basis for the paper's architectural claims, rather than assuming that the complete system outperforms simpler alternatives without testing it.

24. Cross-Country, Cross-Region and Temporal Generalization Experiments

24.1 Country-held-out evaluation

Models are trained on a large subset of the 141 available economies and evaluated on entirely unseen countries, repeated across multiple held-out country selections to assess variance. This tests whether learned predictors and explanations transfer to countries not seen during training.

24.2 Region-held-out evaluation

Models are trained on all regions except one (e.g., trained on Sub-Saharan Africa, South Asia, Latin America and the Caribbean, and Europe and Central Asia; tested on the Middle East and North Africa), repeated across each region in turn. This tests generalization at a coarser, policy-relevant grain.

24.3 Income-group-held-out evaluation

Models are trained on economies from a subset of World-Bank income-group classifications and evaluated on a held-out income group, testing whether financial-inclusion drivers identified in higher-income contexts transfer to lower-income contexts and vice versa — a question of direct relevance to development-finance policy audiences.

24.4 Temporal robustness

Where variable definitions are sufficiently harmonized across editions, models are trained on earlier Global Findex waves (2011–2021) and evaluated on the 2024/2025 wave, testing robustness to changes in financial technology and infrastructure over time — most notably the substantial growth in mobile-money and digital-payment adoption documented across editions.

24.5 Subgroup shift

The same generalization protocol is repeated across gender, income tercile, age cohort and rural/urban residence, using the demographic breakdowns the Global Findex database reports natively, to test whether the model identifies different barriers for different populations and whether its explanations remain stable within versus across those populations.

25. Failure Analysis Taxonomy

A structured failure taxonomy is defined and quantified across the evaluation set, rather than only reporting aggregate success metrics.

- Type A — Incorrect financial-inclusion prediction.
- Type B — Correct prediction but misleading or unfaithful explanation.
- Type C — Valid explanation but an implausible or non-actionable counterfactual.
- Type D — Plausible counterfactual but weak or missing supporting evidence.
- Type E — Adequate evidence but poor personalization to the individual's actual context.
- Type F — Otherwise correct recommendation but overstated causal language (a violation of the causal-claim ladder in Section 19).
- Type G — High system uncertainty that should have triggered abstention but did not.

26. Statistical Analysis Plan and Retrieval Complexity-versus-Benefit Test

All comparisons across model configurations, retrieval strategies and ablation stages use paired statistical tests appropriate to the metric (e.g., paired bootstrap for AUC differences, McNemar's test for classification disagreement, paired Wilcoxon signed-rank for expert-rating differences), reported with effect sizes and

confidence intervals rather than p-values alone. Multiple-comparison correction is applied where the ablation and generalization experiments generate many pairwise comparisons.

Retrieval complexity is explicitly tested rather than assumed beneficial: dense, hybrid and graph-based retrieval are compared on evidence-support quality, latency and computational cost, so that any adoption of the more complex GraphRAG approach is justified by a measured improvement rather than by its novelty.

27. Ethics, Privacy and Responsible-AI Considerations

- Global Findex microdata is anonymized, publicly released survey microdata; no additional individual re-identification risk is introduced by this study, but standard responsible-data-handling practices (secure storage, access logging, no attempted re-identification or linkage to external individual-level sources) are followed.
- Any Tier 3 restricted dataset obtained through direct researcher access will require its own ethical approval, data-use agreement, and, where applicable, institutional review board clearance before analysis begins.
- Demographic variables (gender, age, income group, rural/urban residence) are used only for the theoretically justified subgroup and context-dependence analyses described in Sections 9, 14 and 24, not as inputs that could drive individualized credit or lending decisions.
- The LLM intervention layer is restricted to decision-support and financial-education framing; the verification layer explicitly prohibits autonomous credit-approval/denial language and guaranteed-outcome claims (Section 17.7).
- The causal-claim ladder (Section 19) and causal-identification gate (Section 18) function as ethical as well as statistical safeguards, preventing the system from overstating what an intervention can be expected to achieve for a real individual.

28. Risk Register

Risk	Mitigation
Restricted Tier 3 dataset access denied or delayed	Core paper is fully supported by Tier 1/Tier 2 public data alone (Section 10); Tier 3 is additive, not blocking
Overclaiming causal effects from observational data	Formal causal-identification gate (Section 18) and causal-claim ladder (Section 19), enforced in both analysis and generated text
Generic, non-personalized LLM output	Generation is conditioned on client-specific ML, XAI and counterfactual outputs; verified against the causal-claim ladder
Retrieval architecture complexity unjustified	Explicit complexity-versus-benefit comparison (Section 26); GraphRAG adopted only if it measurably outperforms simpler retrieval
Overclaiming architectural novelty	Novelty positioned as the cross-context generalization study plus the constrained evidence-to-language pipeline (Section 3), not as any single component
Survey-weighting or cross-country	Explicit use of documented Global Findex survey weights and

Risk	Mitigation
harmonization errors	codebook-based variable harmonization; validation against published country-level Findex statistics
Privacy or ethical concerns with any Tier 3 data	De-identification, access control, and ethical approval prior to any restricted-data analysis (Section 27)

29. Success Criteria

30. The primary ML task has a reproducible evaluation protocol, including documented random, country-held-out, region-held-out and income-group-held-out splits.
31. The selected predictive model demonstrates useful discrimination and calibration without unacceptable subgroup degradation.
32. Explanations are demonstrably faithful to model behavior and their cross-context stability is quantified, not assumed.
33. The stable-versus-context-dependent driver classification (Section 14) is produced with a documented, replicable methodology.
34. Counterfactual scenarios satisfy predefined validity, proximity and domain-plausibility constraints; causal estimates, if produced, pass the identification gate with assumptions stated in full.
35. The retrieval component demonstrates measurable evidence-retrieval quality, and any added retrieval complexity is empirically justified.
36. LLM outputs show high evidence support, low unsupported-claim rates, and correct causal-claim-ladder placement.
37. Blinded expert evaluators judge personalized interventions as relevant and appropriate above predefined acceptance thresholds.
38. The ablation study identifies which components materially contribute to final system quality.
39. All experimental configurations, data splits, and evaluation code are reproducible and, where licensing permits, released.

30. Proposed Final System Output

For a given individual record, the system generates a structured intervention report containing:

40. Client-state summary based on permitted data.
41. Predicted financial-inclusion (or selected target) state with calibrated uncertainty.
42. Top contributing factors, with an explicit note on whether each is a globally stable or context-specific driver.
43. Identified barriers or intervention opportunities.
44. Counterfactual or (where identified) causal scenarios, with quantitative estimates, explicit assumptions, and correct causal-claim-ladder labeling.

45. A recommended, evidence-grounded intervention pathway.
46. Supporting financial-inclusion/financial-literacy evidence with source references.
47. Personalized educational or action steps.
48. Stated limitations, uncertainty, and any human-review flag.

31. Publication Strategy and Target Journal Structure

The manuscript is structured to match the expectations of high-impact-factor journals at the intersection of development finance, fintech and applied AI (e.g., outlets publishing financial-inclusion, digital-finance and explainable-AI empirical work), which typically expect a clear theoretical model, a globally relevant and reproducible dataset, a rigorous generalization or heterogeneity analysis, and explicit engagement with causal-inference limitations — all of which this revised design now provides directly.

49. Introduction and motivation
50. Literature review: financial inclusion theory, microfinance and credit-risk ML, explainable ML for financial inclusion, causal ML (including DML), LLMs and retrieval for financial applications
51. Theoretical framework and research hypotheses
52. Dataset and variable construction, including the global public-dataset landscape and selection criteria
53. Proposed MicroFin-XAI framework
54. ML, explainability and counterfactual/causal methodology, including the identification gate and claim ladder
55. Evidence retrieval and LLM intervention methodology
56. Experimental design, baselines and ablation protocol
57. Results: prediction, explanation, generalization (stable vs. context-dependent drivers)
58. Cross-country, cross-region, income-group and temporal generalization results
59. Subgroup, robustness and sensitivity analysis
60. Failure analysis
61. Expert evaluation results
62. Discussion: theoretical and policy implications
63. Limitations and future work (including the Tier 3 causal-validation track, if pursued)
64. Conclusion

32. Reproducibility Plan

- Public release, where licensing and ethics permit, of preprocessing scripts, variable-mapping/harmonization code, and country/region/income-group split definitions.
- Fixed random seeds, documented model configurations and hyperparameter search spaces for every reported result.

- Versioned evaluation scripts for predictive, explanation-faithfulness, counterfactual-validity, retrieval and verification metrics.
- Versioned retrieval-corpus metadata (source list, collection date, licensing) without redistributing copyrighted full texts where restricted.
- Documented prompts and verification rules for the LLM layer.
- Documented expert-evaluation protocol, rating instrument, and statistical-analysis code.
- For any Tier 3 restricted data used in the causal-validation track, release of derived metadata and analysis code only, consistent with the originating data-use agreement.

33. Collaboration Roles

Area	Primary expertise
Financial-inclusion theory, constructs, hypotheses	Domain researcher / supervisor
Survey/construct validation and variable harmonization	Domain researcher + collaboration
Dataset acquisition, cleaning and interpretation	Joint
ML pipeline and benchmarking	AI/ML researcher
XAI, counterfactual and causal-gate implementation	AI/ML researcher + methodological review
Evidence corpus and retrieval systems	AI/GenAI researcher
LLM intervention and verification layer	AI/GenAI researcher
Domain validity and blinded expert evaluation	Joint
Statistical interpretation and manuscript framing	Joint
Reproducibility and software artifact release	AI/ML researcher

34. Conclusion

This revised proposal repositions MicroFin-XAI from a single-country, access-dependent microfinance system concept into a globally grounded, reproducible, access-realistic research design without discarding the microfinance domain expertise that motivated the original proposal. The Global Findex Database 2025 supplies a public, individual-level, cross-country dataset sufficient to carry the paper's central empirical contribution — the identification of globally stable versus context-dependent financial-inclusion barriers under a rigorously tested distribution-shift protocol — while the original microfinance-specific datasets are retained as a secondary, access-pending causal-validation track rather than a single point of failure.

The framework's scientific core is not the number of AI components it contains, but the discipline with which it separates association, prediction, model-based counterfactual reasoning, and (where identifiable) causal estimation, and the rigor with which every generated statement is held to the evidentiary level it has actually earned. That discipline, combined with a genuinely global and reproducible empirical foundation, is what

positions MicroFin-XAI for a high-impact-factor (Q1) journal contribution, rather than an ambitious but under-grounded AI system proposal.

As with the original proposal, development should proceed only after the primary outcome definition, the finalized variable taxonomy, and the causal-identification scope are agreed upon jointly with the domain researcher, using the concrete next steps below.

Immediate Next Steps

(1) Download and complete a full variable-by-variable audit of the Global Index 2025 codebook and questionnaire. (2) Fix the primary outcome definition with the domain researcher (Section 11.2). (3) Submit access requests for the Tier 3 datasets (Section 10.3) in parallel, without blocking the core timeline on their approval. (4) Draft the theoretical framework and hypotheses as a standalone pre-registration-style document before any model training begins.